

Homework - III

Statistics III : Multivariate Data and Regression

Utkarsh Bharadwaj

Roll Number: BSD-BG-2521

October 1, 2026

Question 1

Consider two independent samples from two normal populations with the same variance.

- Set up the two-sample t -test for testing the equality of means of the two populations.
- Set up the same problem as a simple linear regression with testing for zero slope.
- Show that the test statistics and their distributions under the null are the same in the two cases.

Solution 1

- Let $X_1, \dots, X_{n_1} \stackrel{iid}{\sim} N(\mu_1, \sigma^2)$ and $Y_1, \dots, Y_{n_2} \stackrel{iid}{\sim} N(\mu_2, \sigma^2)$ be independent samples. We want to test $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 \neq \mu_2$. The pooled variance estimator is

$$s_p^2 = \frac{\sum_{i=1}^{n_1} (X_i - \bar{X})^2 + \sum_{j=1}^{n_2} (Y_j - \bar{Y})^2}{n_1 + n_2 - 2}$$

and the test statistic is

$$T = \frac{\bar{Y} - \bar{X}}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2} \text{ under } H_0$$

- Let $N = n_1 + n_2$ and stack the observations as $y_1, \dots, y_N$. Define the indicator $z_i = 0$ if y_i is from the first sample and $z_i = 1$ if it is from the second sample. Consider the model

$$y_i = \beta_0 + \beta_1 z_i + \varepsilon_i, \quad \varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$$

Then $E(y_i) = \beta_0 = \mu_1$ for the first sample and $E(y_i) = \beta_0 + \beta_1 = \mu_2$ for the second sample, so $\beta_1 = \mu_2 - \mu_1$. Thus, testing $\mu_1 = \mu_2$ is the same as testing $H_0 : \beta_1 = 0$.

- We have $\bar{z} = n_2/N$. Thus, we have the following

$$S_{zz} = \sum_{i=1}^N (z_i - \bar{z})^2 = n_1 \left(\frac{n_2}{N}\right)^2 + n_2 \left(\frac{n_1}{N}\right)^2 = \frac{n_1 n_2}{N}$$

$$S_{zy} = \sum_{i=1}^N (z_i - \bar{z}) y_i = \frac{n_1}{N} n_2 \bar{y}_2 - \frac{n_2}{N} n_1 \bar{y}_1 = \frac{n_1 n_2}{N} (\bar{y}_2 - \bar{y}_1)$$

where $\bar{y}_1 = \bar{X}$ and $\bar{y}_2 = \bar{Y}$. This implies

$$\hat{\beta}_1 = \frac{S_{zy}}{S_{zz}} = \bar{Y} - \bar{X}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{z} = \bar{X}$$

So the fitted values are simply the group means, i.e., $\hat{y}_i = \bar{X}$ for the first sample and $\hat{y}_i = \bar{Y}$ for the second sample. Hence

$$SSRes = \sum_{i=1}^{n_1} (X_i - \bar{X})^2 + \sum_{j=1}^{n_2} (Y_j - \bar{Y})^2 \implies s^2 = \frac{SSRes}{N-2} = s_p^2$$

Also, $\frac{1}{S_{zz}} = \frac{N}{n_1 n_2} = \frac{1}{n_1} + \frac{1}{n_2}$. Thus, the regression t -statistic is

$$t = \frac{\hat{\beta}_1}{s/\sqrt{S_{zz}}} = \frac{\bar{Y} - \bar{X}}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = T$$

Under $H_0 : \beta_1 = 0$, we know from simple linear regression that $t \sim t_{N-2} = t_{n_1+n_2-2}$. Thus, both the statistic and its null distribution are identical in the two cases.

Question 2

Consider the `seatpos` data. Carry out a model selection exercise. Present all details in intermediate steps, the final model with proper interpretations of output.

Solution 2

The response is `hipcenter` with eight candidate predictors and $n = 38$. We select the model using adjusted R^2 , AIC, SBC and Mallows' C_p (all computed by `olsrr`). A model is better if it has a larger adjusted R^2 , smaller AIC and SBC, and C_p close to (or below) p , the number of parameters including the intercept.

(a) **Full model.** Consider the following fit.

Listing 1: Full model for `seatpos`

```
1 library(readxl)
2 library(olsrr)
3 seatpos <- na.omit(read_excel("seatpos.xlsx"))
4
5 full <- lm(hipcenter ~ Age + Weight + HtShoes + Ht +
6           Seated + Arm + Thigh + Leg, data = seatpos)
7 summary(full)
```

We get $R^2 = 0.687$ and adjusted $R^2 = 0.600$. The overall F -test is significant ($p = 1.31 \times 10^{-5}$) but no predictor is individually significant (smallest p -value is 0.182 for `Leg`). As seen in Homework 2, the size-related predictors are highly correlated (e.g., `Ht` and `HtShoes` at 0.998), so some predictors are redundant and we should look for a smaller model.

(b) **All possible subsets.** There are $2^8 - 1 = 255$ candidate models. We compute all four criteria for each of them.

Listing 2: All possible regressions using `olsrr`

```
1 all_models <- ols_step_all_possible(full)
2 all_models[, c("n", "predictors", "adjr", "cp", "aic",
3               , "sbc")]
4
5 # best model according to each criterion
6 all_models[which.max(all_models$adjr), ]
7 all_models[which.min(all_models$aic), ]
8 all_models[which.min(all_models$sbc), ]
9 all_models[which.min(all_models$cp), ]
```

Criterion	Best model	Value
Adjusted R^2 (max)	Age, Ht, Leg	0.6533
AIC (min)	Age, Ht, Leg	274.24
SBC (min)	Ht	278.34
C_p (min)	Ht	-0.53

Table 1: Best model among all 255 subsets for each criterion.

- (c) **Best model of each size.** To see the trade-off more clearly, consider the best model for each number of predictors.

Listing 3: Best subset of each size

```

1 best <- ols_step_best_subset(full)
2 best
3 plot(best)

```

Size	Best subset	Adj. R^2	C_p	AIC	SBC
1	Ht	0.6282	-0.53	275.07	278.34
2	Ht, Leg	0.6399	-0.49	274.78	279.69
3	Age, Ht, Leg	0.6533	-0.52	274.24	280.79
4	Age, HtShoes, Thigh, Leg	0.6467	1.16	275.83	284.02
5	Age, HtShoes, Arm, Thigh, Leg	0.6371	3.04	277.67	287.50
6	Age, HtShoes, Seated, Arm, Thigh, Leg	0.6257	5.01	279.64	291.10
7	all except Ht	0.6134	7.00	281.63	294.73
8	all	0.6001	9.00	283.62	298.36

Table 2: Best subset of each size ($p = \text{size} + 1$).

We observe the following:

- Adjusted R^2 and AIC both peak at size 3, with the model {Age, Ht, Leg}.
- C_p is below p for sizes 1 to 3, and from size 4 onwards it is close to p with no gain in fit. Thus, models of size ≤ 3 are adequate.
- SBC penalizes the model size more heavily and prefers the single predictor Ht, but its value for size 3 is only slightly larger.
- Adjusted R^2 and AIC are nearly the same for {Age, Ht, Leg} and {Age, HtShoes, Leg} (AIC 274.24 vs 274.26) since Ht and HtShoes are almost identical.

- (d) **Final model.** Three of the four criteria (adjusted R^2 , AIC and C_p in the sense $C_p < p$) support {Age, Ht, Leg}, which is also the best model of size 3 overall. Thus, we select it.

Listing 4: Final model and diagnostics

```

1 final <- lm(hipcenter ~ Age + Ht + Leg, data =
2   seatpos)
3 summary(final)
4
5 ols_plot_resid_fit(final)
6 ols_plot_resid_qq(final)
7 ols_plot_resid_hist(final)

```

```
7 ols_plot_resid_lev(final)
```

	Estimate	Std. Error	t value	p-value
Intercept	452.198	100.948	4.480	< 0.001
Age	0.581	0.379	1.532	0.135
Ht	-2.325	1.254	-1.854	0.072
Leg	-6.739	4.105	-1.642	0.110

Table 3: Coefficients of the final model. $R^2 = 0.681$, adj. $R^2 = 0.653$, $\hat{\sigma} = 35.12$, $F_{3,34} = 24.24$ ($p = 1.43 \times 10^{-8}$).

The adjusted R^2 rises from 0.600 (full model) to 0.653 while R^2 only drops from 0.687 to 0.681. Thus, the five dropped predictors added almost nothing.

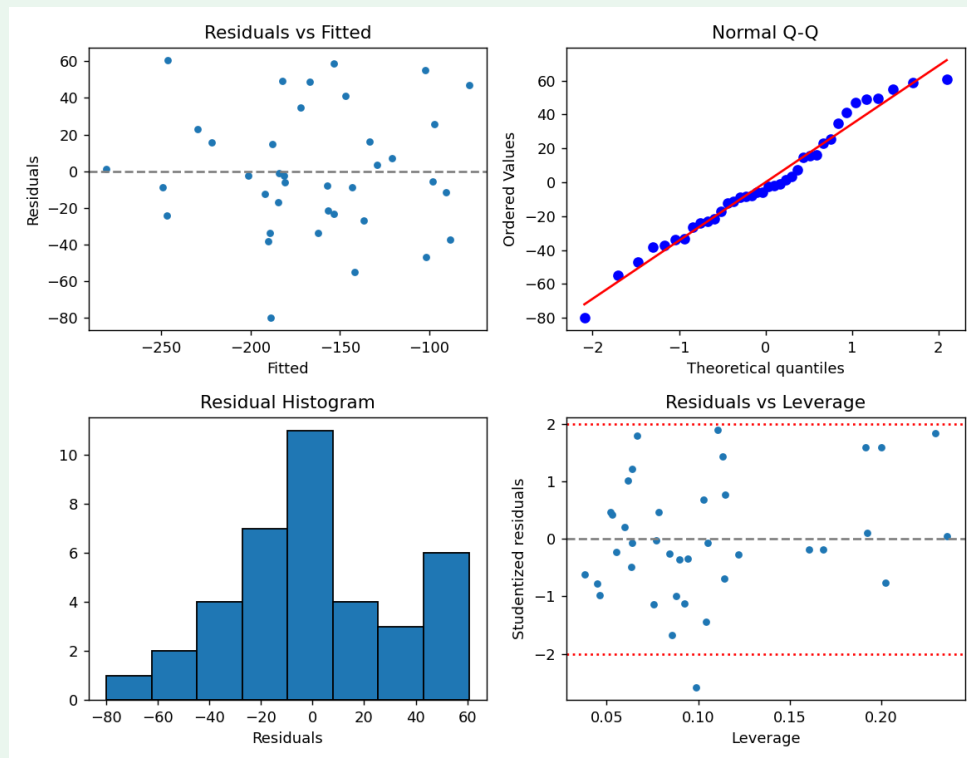


Figure 1: Diagnostic plots of residuals for the final model.

The residuals have no visible pattern against the fitted values, the Q-Q plot is roughly linear and the histogram is roughly symmetric, so the linearity, constant variance and normality assumptions look reasonable. The maximum leverage is $0.236 < 3p/n = 0.316$ and the maximum Cook's distance is 0.24, so no observation is highly influential (only one studentized residual is slightly above 2 in absolute value).

(e) **Interpretation.** The fitted model is

$$\widehat{\text{hipcenter}} = 452.20 + 0.58 \text{ Age} - 2.33 \text{ Ht} - 6.74 \text{ Leg}$$

Holding the other predictors fixed:

- each additional year of age increases `hipcenter` by about 0.58 units (older drivers sit slightly further forward),
- each additional cm of height decreases `hipcenter` by about 2.33 units, i.e., taller drivers sit further back,
- each additional cm of leg length decreases `hipcenter` by about 6.74 units, i.e., drivers with longer legs sit further back.

The model explains 68.1% of the variation in `hipcenter` and the overall F -test is highly significant. Individually, none of the three coefficients is significant at 5% (p -values 0.072 to 0.135) because `Ht` and `Leg` are still strongly correlated (0.910) and n is small. Thus, their separate effects cannot be pinned down sharply, but jointly they are clearly important. If a simpler model is preferred, `Ht` alone (chosen by SBC and C_p) already gives $R^2 = 0.638$.

Question 3

A single factor completely randomized design has six levels of the factor. There are five replicates at each level and the total sum of squares is 900.25. The treatment sum of squares is 750.5.

- What is the estimate of the error variance?
- What proportion of variability is explained by the treatment effect?
- What is the F -statistic and corresponding p -value?

Solution 3

Here $k = 6$ and $N = 6 \times 5 = 30$. Thus, $SSTr = 750.5$ and $SSE = SST - SSTr = 900.25 - 750.5 = 149.75$. The degrees of freedom are $k - 1 = 5$ for treatment and $N - k = 24$ for error.

- The estimate of the error variance is

$$\hat{\sigma}^2 = MSE = \frac{SSE}{N - k} = \frac{149.75}{24} \approx 6.24$$

- The proportion of variability explained is

$$R^2 = \frac{SSTr}{SST} = \frac{750.5}{900.25} \approx 0.8337$$

i.e., about 83.4%.

- We have $MSTr = \frac{750.5}{5} = 150.1$. Thus, the F -statistic is

$$F = \frac{MSTr}{MSE} = \frac{150.1}{6.24} \approx 24.06$$

Under H_0 : all level means are equal, $F \sim F_{5,24}$. Thus, the p -value is

$$p = P(F_{5,24} > 24.06) \approx 1.27 \times 10^{-8}$$

Since the p -value is very small, we reject H_0 , i.e., the treatment effect is significant.

Question 4

Consider an experiment with continuous response (y) and one factor (x) with k levels. Suppose there are n_i independent measurements at each level for $i = 1, \dots, k$.

- Write down the linear model and explain the problem of non-identifiability.
- If we want to obtain point estimates of the level means, then what should be the procedure for removing non-identifiability? What is the new design (X) matrix?
- In the set-up of (b) find the estimate of the vector of level means.
- Find the covariance matrix of the estimate obtained in part (c).
- Assuming independent normal additive errors, estimate the error variance.

Solution 4

Let $N = \sum_{i=1}^k n_i$.

- Consider the following model

$$y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i = 1, \dots, k, \quad j = 1, \dots, n_i$$

with $E(\varepsilon_{ij}) = 0$ and $Var(\varepsilon_{ij}) = \sigma^2$. In matrix form $Y = X\beta + \varepsilon$ where $\beta = (\mu, \alpha_1, \dots, \alpha_k)^T$ and X is $N \times (k+1)$. The first column of X (intercept) is the sum of the remaining k indicator columns, so $\text{rank}(X) = k < k+1$ and $X^T X$ is singular. Equivalently, for any constant c , the parameters $(\mu + c, \alpha_1 - c, \dots, \alpha_k - c)$ give the same $E(Y)$. Thus, β is not identifiable and the normal equations have infinitely many solutions. Only the level means $\mu_i = \mu + \alpha_i$ are identifiable.

- Since we only want the level means, we remove the intercept (equivalently, impose the constraint $\mu = 0$) and fit the cell means model

$$y_{ij} = \mu_i + \varepsilon_{ij}$$

The new design matrix is the $N \times k$ matrix

$$X = \begin{pmatrix} \mathbf{1}_{n_1} & 0 & \cdots & 0 \\ 0 & \mathbf{1}_{n_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathbf{1}_{n_k} \end{pmatrix}$$

which has full column rank k .

- We have $X^T X = \text{diag}(n_1, \dots, n_k)$ and $X^T Y = (n_1 \bar{y}_1, \dots, n_k \bar{y}_k)^T$ where $\bar{y}_i = \frac{1}{n_i} \sum_j y_{ij}$. Thus, we have

$$\hat{\mu} = (X^T X)^{-1} X^T Y = (\bar{y}_1, \dots, \bar{y}_k)^T$$

- We know that $Var(\hat{\mu}) = \sigma^2 (X^T X)^{-1}$. This implies

$$Var(\hat{\mu}) = \sigma^2 \text{diag} \left(\frac{1}{n_1}, \dots, \frac{1}{n_k} \right)$$

i.e., the estimates of the level means are uncorrelated.

(e) The residuals are $y_{ij} - \hat{y}_{ij} = y_{ij} - \bar{y}_i$. Thus, we have

$$SSRes = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2$$

Since $\text{rank}(X) = k$, we have $SSRes/\sigma^2 \sim \chi_{N-k}^2$. Hence the unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{SSRes}{N-k} = \frac{1}{N-k} \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2$$